

Module 2 Lab 3 - Textbook Problem 4.4: Temperature Categories

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GitHub Repo: <https://github.com/ramishatasfia5156-commits/cs82a-portfolio/tree/main/module2>

Temperature Cutoff Rules

All temperatures are in degrees Celsius ($^{\circ}\text{C}$), recorded over a week.

Category	Rule
Cold	temperature < 15°C
Warm	$15^{\circ}\text{C} \leq \text{temperature} \leq 25^{\circ}\text{C}$
Hot	temperature > 25°C

Boundary rule: 15°C is *Warm* (inclusive lower bound); 25°C is *Warm* (inclusive upper bound); anything strictly above 25°C is *Hot*.

```
In [1]: # Problem 4.4 – Temperature Categories
# Categorizes each day's temperature as Cold, Warm, or Hot.

Temperatures = [10, 16, 23, 28, 14, 22, 30]

print(f"{'Temperature':>13}  {'Category'}")
print("-" * 26)

for temp in Temperatures:
    if temp < 15:
        category = "Cold"
    elif temp <= 25:
        category = "Warm"
    else:
        category = "Hot"

    print(f"{temp:>11}°C  {category}")
```

Temperature	Category
10°C	Cold
16°C	Warm
23°C	Warm
28°C	Hot
14°C	Cold
22°C	Warm
30°C	Hot

Boundary Case Verification

Testing the exact boundary values (14, 15, 25, 26) to confirm the `elif` chain matches the stated cutoff rules.

```
In [2]: # Boundary spot-check
boundary_tests = [
    (14, "Cold"),
    (15, "Warm"),
    (25, "Warm"),
    (26, "Hot"),
]

print("Boundary verification:")
print(f"{'Temp':>6}  {'Expected':>8}  {'Got':>8}  {'Pass?'}")
print("-" * 36)

all_pass = True
for temp, expected in boundary_tests:
    if temp < 15:
        got = "Cold"
    elif temp <= 25:
        got = "Warm"
    else:
        got = "Hot"

    result = "PASS" if got == expected else "FAIL"
    if got != expected:
        all_pass = False
    print(f"{temp:>5}°C  {expected:>8}  {got:>8}  {result}")

print()
print("All boundary cases passed!" if all_pass else "One or more boundary ca")
```

Boundary verification:

Temp	Expected	Got	Pass?
14°C	Cold	Cold	PASS
15°C	Warm	Warm	PASS
25°C	Warm	Warm	PASS
26°C	Hot	Hot	PASS

All boundary cases passed!

AI Disclosure: Claude by Anthropic was used to help draft and explain concepts in this lab. All logic, cutoff decisions, and final code were reviewed and verified by me, Ramisha Tasfia.